

Akash Sureshkumar

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SUMMARY

Software Engineer with 3+ years building full-stack tools, REST APIs, automation workflows, and production systems using Java, Python, JavaScript/TypeScript, React, Node.js, SQL, Docker, Linux, and AWS. Strong in CI/CD, test automation, data modeling, stakeholder-driven tooling, scalable backend systems, and AI-assisted developer workflows.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, Bash

Full Stack: React, Angular, Node.js, Spring Boot, REST APIs, Microservices, JWT/OAuth2, HTML/CSS

Automation & Quality: Test Automation, API Testing, UI Workflows, JUnit, Mockito, Postman, CI/CD, GitHub Actions

Cloud & Data: AWS, Docker, Linux, CloudWatch, Redis, PostgreSQL, MySQL, Oracle, Splunk, AppDynamics

Engineering: Internal Tools, Data Modeling, Object-Oriented Design, SDLC, Code Review, Agile/Scrum, AI-Assisted Workflows

EXPERIENCE

University at Buffalo

Buffalo, NY

Software Engineer, Research

Feb 2026 – Present

- Built a full-stack Faculty Portal for 200+ users across 6 departments by delivering 7+ Java Spring Boot REST APIs, JWT/OAuth2 authentication, PostgreSQL data models, and reusable React/Next.js workflows.
- Reduced average query response time by 40% and database read load by 35% by designing normalized schemas, adding index-aware SQL patterns, and optimizing Hibernate/JPA queries across 8+ workflows.
- Reduced deployment cycle time from 30 minutes to under 5 minutes by containerizing services with Docker and implementing GitHub Actions CI/CD with automated test gates on Linux infrastructure.

Guardian Life

Chennai, India

Software Engineer II

Apr 2024 – Aug 2024

- Reduced MTTR from 5 days to 2 days for insurance benefits workflows at Guardian Life, which serves 12M+ people across life, disability, dental, vision, and accident insurance, by building Splunk dashboards and AWS CloudWatch alarms.
- Reduced release cycle time from 1 hour to 15 minutes and improved shipping speed 3x by helping decompose a monolith into event-driven microservices with isolated CI/CD pipelines.
- Improved delivery quality across 4 Agile teams by clarifying requirements with product owners and QA, scoping backend deliverables, and shipping maintainable eligibility workflow features.
- Reduced cyclomatic complexity by 75% across 25+ services and improved PR throughput by 20% by enforcing SonarQube quality gates, conducting code reviews, and mentoring 7 engineers.

Guardian Life

Chennai, India

Software Engineer I

Apr 2022 – Mar 2024

- Reduced API p95 latency from 455 ms to 120 ms and improved throughput 3x by rewriting Oracle/SQL execution plans, adding covering indexes, and introducing Redis caching with TTL-based invalidation.
- Improved system reliability by 35% and reduced critical incidents by 40% by implementing circuit breaker patterns and automated Postman regression suites across 20+ REST API contracts.
- Reduced manual workflow time by 15% per feature by delivering 5+ Java Spring Boot microservice features for high-volume insurance eligibility and transaction processing workflows.
- Reduced new-engineer onboarding time from 2 days to 4 hours by building 3+ Angular workflows with reusable service layers and authoring setup, database, deployment, and troubleshooting runbooks.

PROJECTS

Pump Fault Risk Prediction Service | *FastAPI, LightGBM, CLIP, Docker, Streamlit* | GitHub

2025

- Built a multimodal predictive service that estimates pump failure risk from real-time sensor telemetry and optional inspection images using FastAPI REST endpoints, health checks, OpenAPI docs, and a Streamlit UI.
- Integrated structured sensor features with CLIP ViT-B/32 image embeddings, LightGBM models, and fusion logic to support calibrated multimodal inference in a Dockerized deployment environment.

Agentic Browser Automation Platform | *TypeScript, React, Node.js, Playwright/Webwright, LLM APIs* | GitHub

2026

- Built an agentic browser automation project with a local-runner Webwright runtime, combining DOM-based
- Added reusable runtime abstractions for browser actions, page-state reasoning, and fallback handling to improve reliability across dynamic pages and complex UI flows. interaction, browser tool orchestration, and visual-fallback execution to automate multi-step web workflows.
- Designed AI-assisted task completion for testing and form workflows by coordinating browser actions with runtime tooling and LLM planning logic.

EDUCATION

University at Buffalo

Buffalo, NY

Master of Science, Engineering Science

Dec 2025

- Coursework: Cloud Computing, Distributed Systems & Big Data, Software Architecture, Machine Learning

Sri Sairam Engineering College

Chennai, India

Bachelor of Engineering, Computer Science and Engineering

Jun 2022

- Coursework: Data Structures & Algorithms, OOP, Software Engineering, Design Patterns, Database Systems